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BLACK PANTHER

Mirrorme Technology

BLACK PANTHER

BPX Assembly Manual



I. Safety Instructions

Dear Engineer, Congratulations on Owning BLACK PANTHER X!

This robotic dog features 4 legs and 12 powerful joints; it transforms from a pile of parts into a running and jumping companion—all thanks to your focused hands-on assembly process!

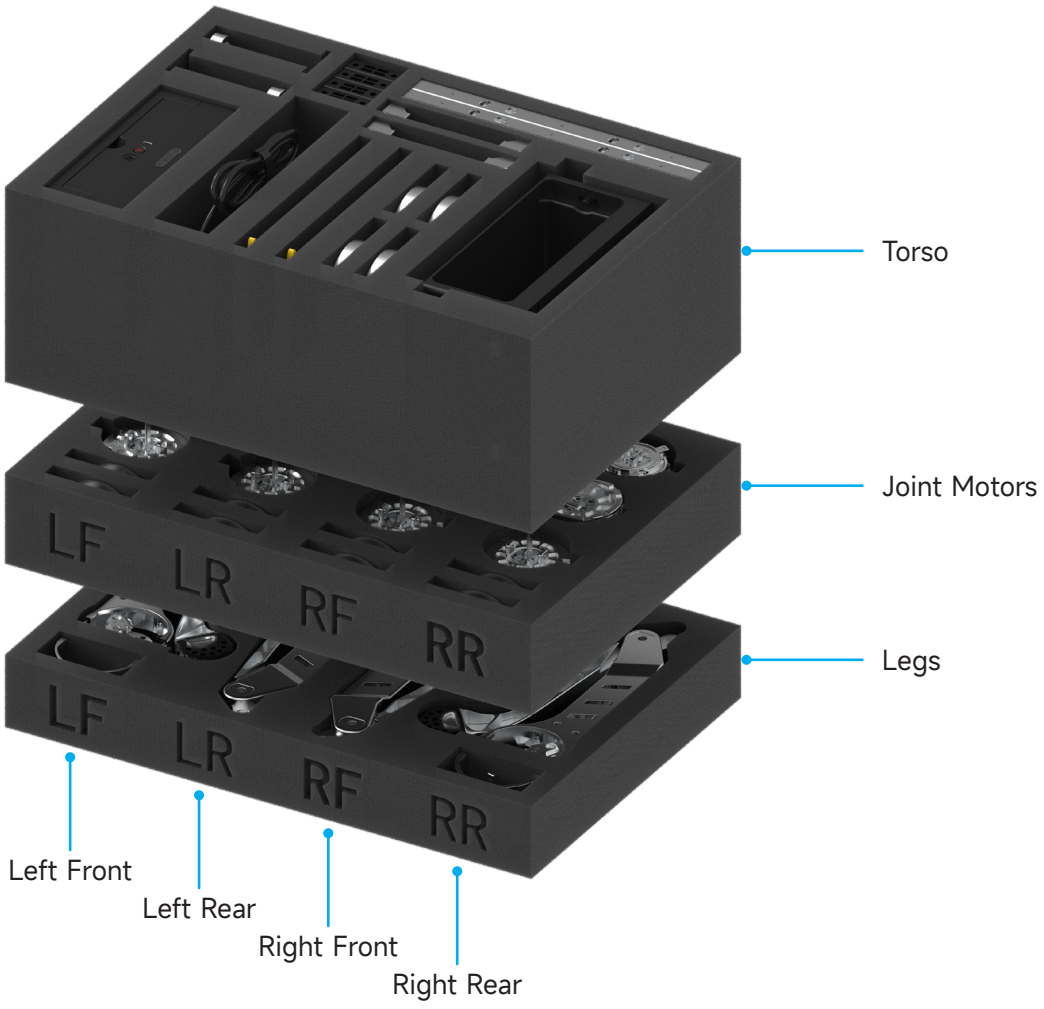
【Safety Warning】

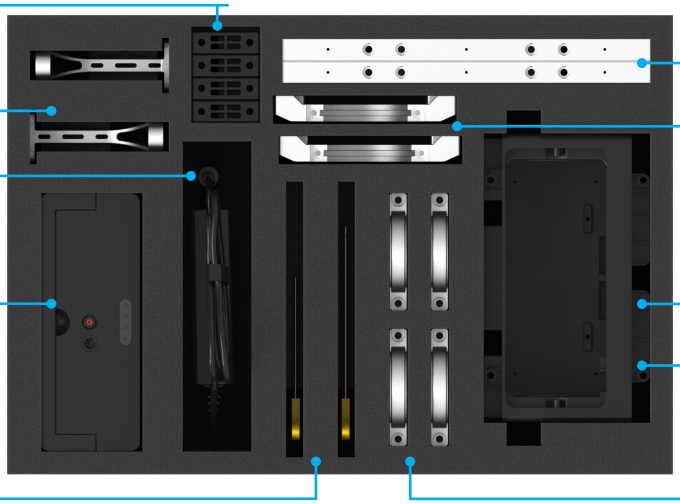
- Children under 12 year old should complete the assembly and power-on process under the supervision of their teacher or parent.
- Do not put small parts in your mouth; avoid the risk of swallowing.
- After turning on the device, please keep your fingers, hair, and cables away from the rotating joint gap to avoid any pinching injuries!

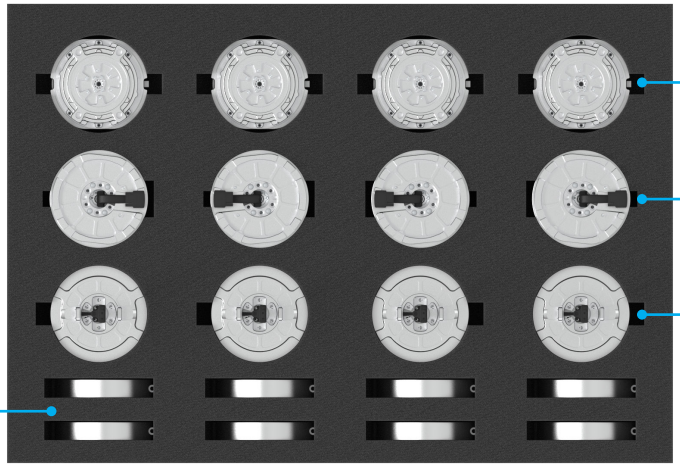
II. Parts display

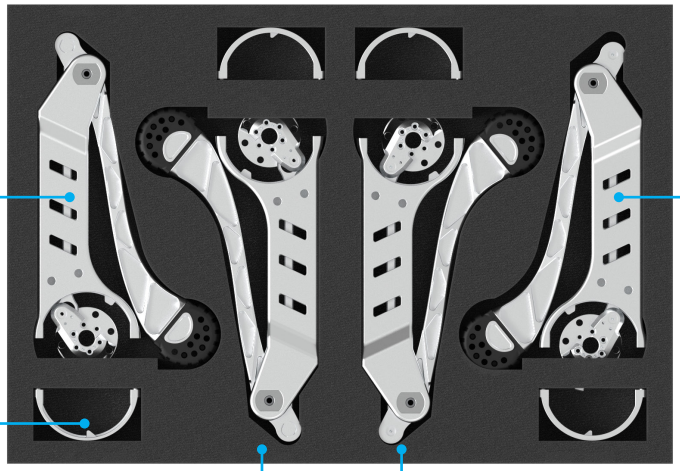
After unboxing, please first check whether all components are complete and neatly arranged. The components inside the box are divided into three layers: the first layer contains the torso components, the second layer contains the joint motors, and the third layer contains the leg components; each layer is designated for placing the corresponding components and tools.

Each side of the layer is printed with position markers:LF (left-front), LR (left-rear), RF (right-front), and RR (right-rear); each column label corresponds to the corresponding leg component.




Bumper Pads
Handles
Battery Charger
Battery
Screwdrivers
Long Beams
& Short Beams
Fixation Base of Beams
Control Box
4× Screws
& Accessories
Torso Hoops


Joint Hoops
Knee Joints
Hip Joints
Side Swing Joints


Left-front Leg
Leg Hoops
Left-rear Leg
Right-rear Leg
Right-front Leg

III.SN Code Encoding Rules

Each motor is labeled with an SN code PET sticker; during assembly, you only need to check the 8 middle letters (positions 3–10). The first 4 letters represent the joint type (item code), while the last 4 letters represent the installation location (sub-item code).

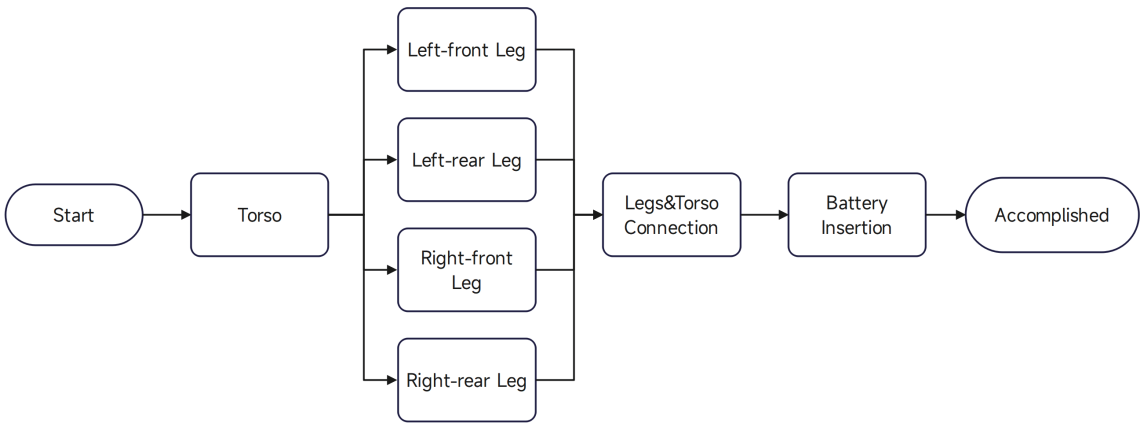
Example:



First 4 Digits	Joint Type	Last 4 Digits	Position
HIPX	Side Swing Joint	LFRR	Left Front / Right Rear
		LRRF	Right Front / Left Rear
HIPY	Hip Joint	LFXX	Left Front
		LRXX	Left Rear
		RFXX	Right Front
		RRXX	Right Rear
KNEE	Knee joint	0STD / 0PRO	Universal

IV. Start Assembling

Sequence Overview

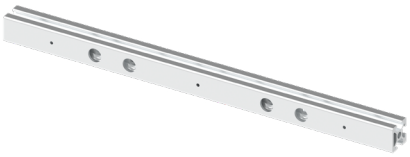


Assembly Instructions:

- For assemblies with multiple screws: loosely fit all screws first. After verifying proper part alignment, tighten them sequentially in diagonal order.

1. Assemble the Torso Frame + Control Box

You will need:



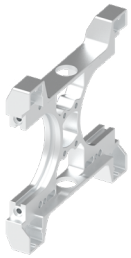
4× Long Beams



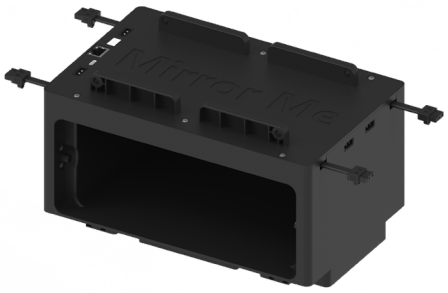
2× Short Beams



4× Bumper Pads



2× Fixation Bases



1× Control Box



16× M6*16 Screws



12× M6*22 Screws

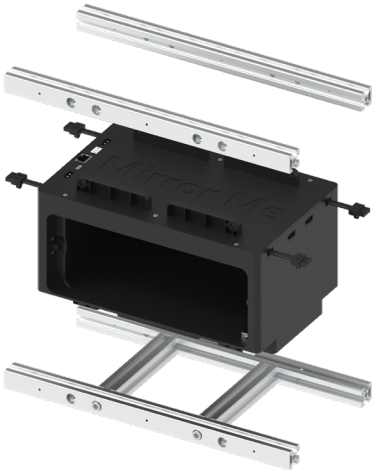
Assembly Method

①



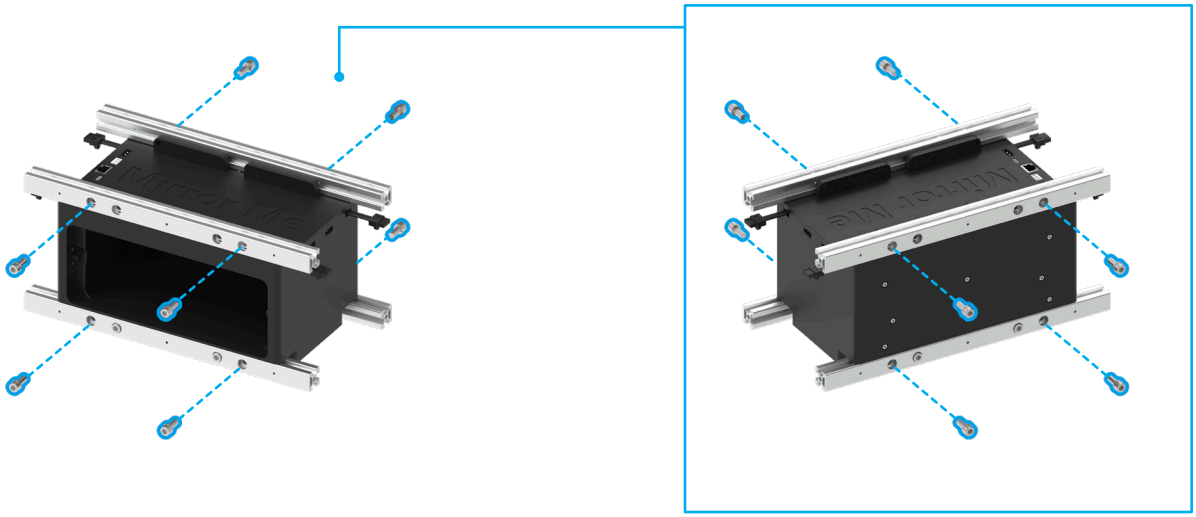
First, take 2 long beams, 2 short beams, and 4 M6*22 screws to assemble the bottom part of the torso frame

②



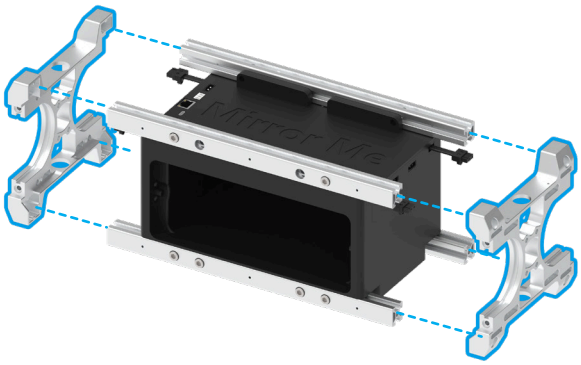
Put the control box into the bottom frame

③

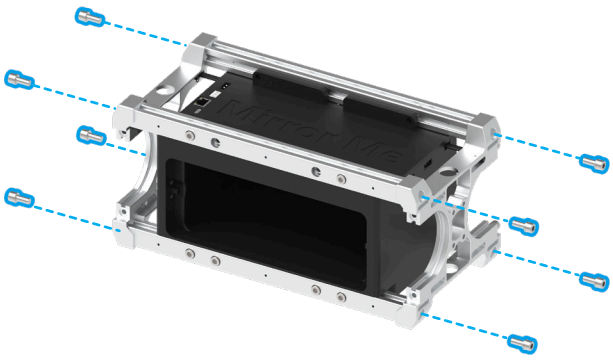


Fix with 8 M6*22 screws

④



Put 2 fixation bases onto the torso frame

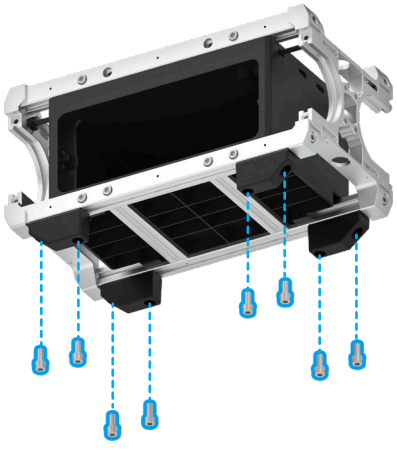


Fix with 8 M6*16 screws

⑤



Put 4 bumper pads onto the torso frame



Fix with 8 M6*16 screws



Accomplished!

2. Assemble Four Legs

You will need:

Left-front Leg	Left-rear Leg	Right-front Leg	Right-rear Leg
1 Leg (L)	1 Leg (L)	1 Leg (R)	1 Leg (R)
1 Knee Joint (KNEE)	1 Knee Joint (KNEE)	1 Knee Joint (KNEE)	1 Knee Joint (KNEE)
1 Hip Joint (HIPYLFXX)	1 Hip Joint (HIPYLRXX)	1 Hip Joint (HIPYRFXX)	1 Hip Joint (HIPYRRXX)
1 Side-swing Joint (HIPXLFRR)	1 Side-swing Joint (HIPXLRRF)	1 Side-swing Joint (HIPXLRRF)	1 Side-swing Joint (HIPXLFRR)
1 Leg Hoop（L）	1 Leg Hoop（L）	1 Leg Hoop（R）	1 Leg Hoop（R）
10 M3*8 Screws	10 M3*8 Screws	10 M3*8 Screws	10 M3*8 Screws
2 M6*16 Screws	2 M6*16 Screws	2 M6*16 Screws	2 M6*16 Screws

Assembly Instructions:

- Legs are distinguished as left and right: the left leg bears the steel stamp L，while the right leg bears the steel stamp R.
- The output shafts of the two front leg side-swing motors are oriented forward.
- The output shafts of the two rear leg side-swing motors are oriented backward.

2.1 Assemble the Left Front Leg

You will need:



1× Left Leg



Knee Joint
(KNEE)



Hip Joint
(HIPYLFXX)



Side-swing Joint
(HIPXLFRR)



Leg Hoop（L）



Knee Hoop

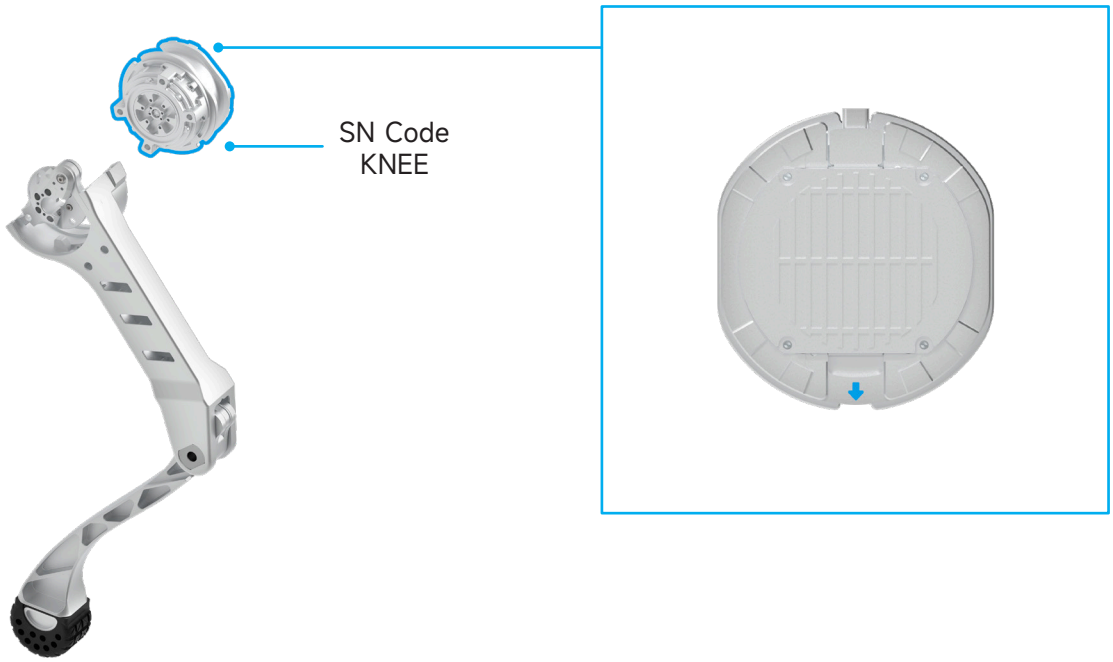
13× M3*8 Screws



2× M6*16 Screws

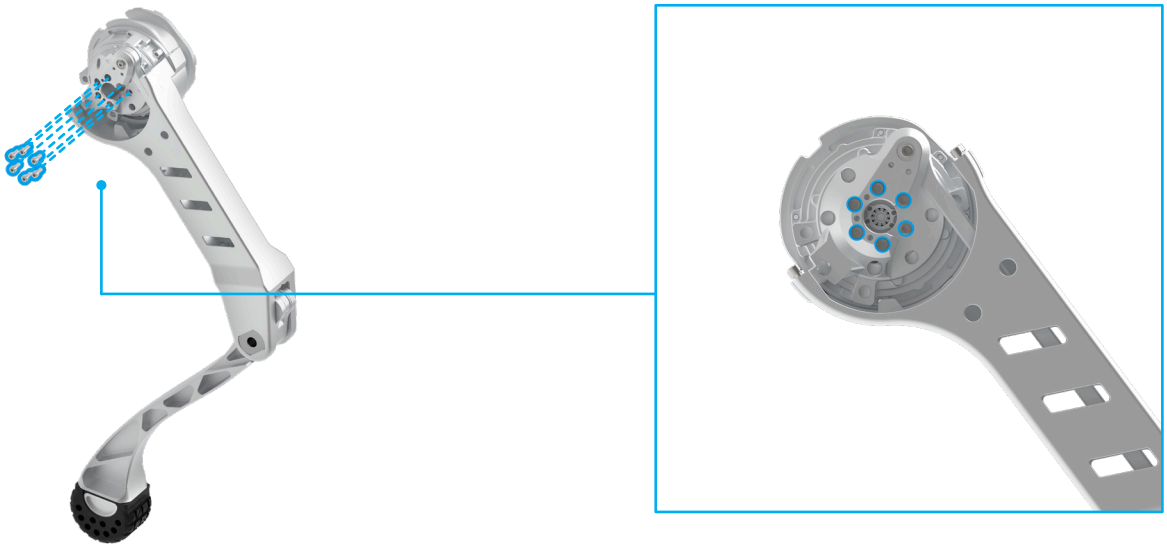
Start Assembling

①



Insert the knee joint into the mounting slot.

②



③



Fix with 6 M3*8 screws

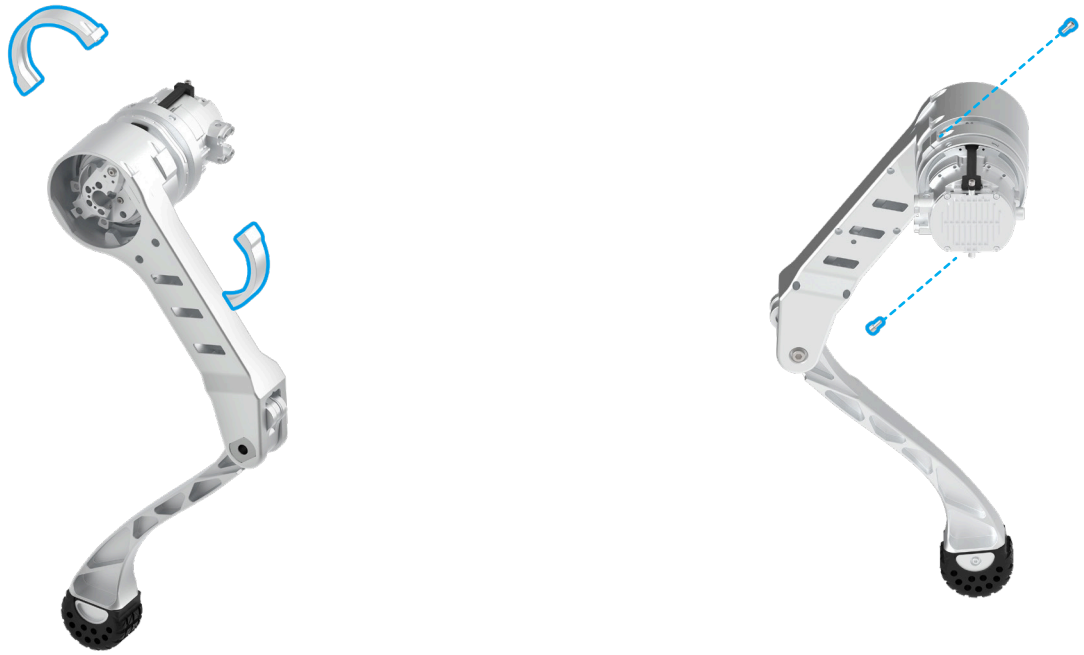
Place the leg hoop over the knee joint and secure it with 4 M3*8 screws

④



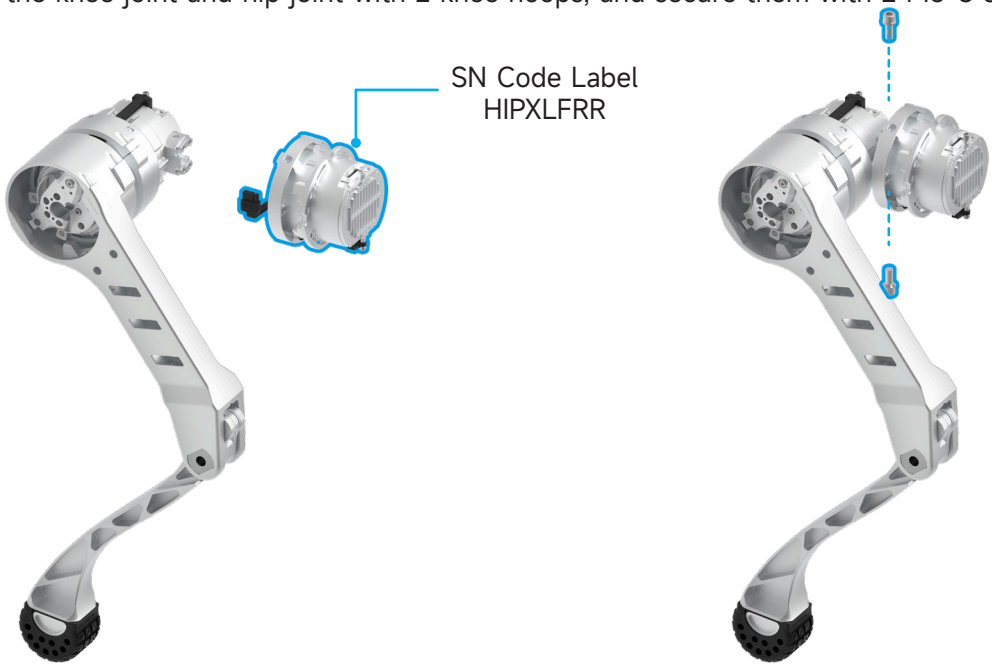
Fix the knee joint and hip joint, connect the cable

⑤

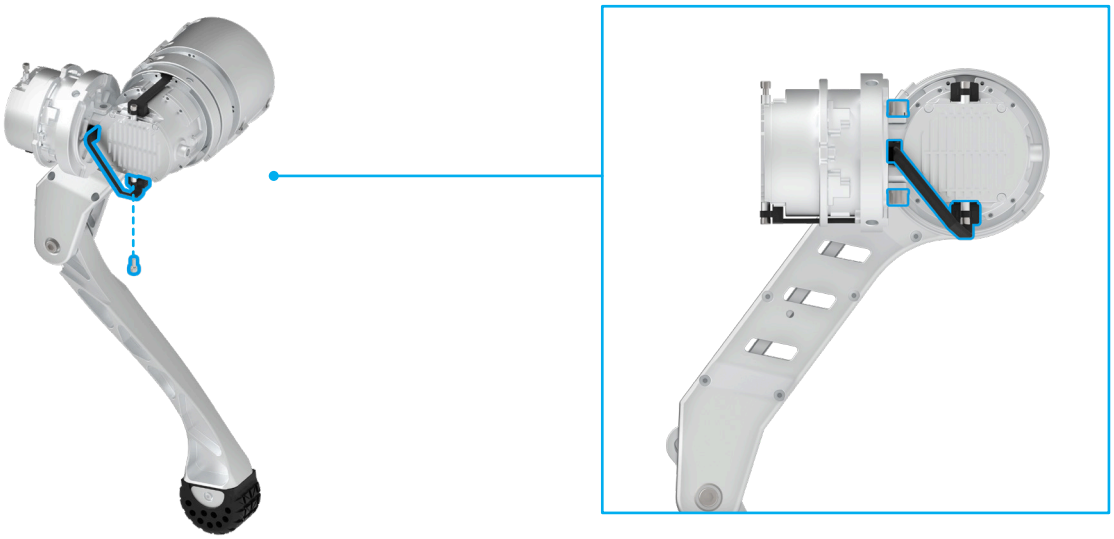


Fix the knee joint and hip joint with 2 knee hoops, and secure them with 2 M3*8 screws

⑥



Align the side-swing joint mounting hole with the hip joint mounting hole and secure with 2 M6*16 screws
* ROUTE THE MOTOR CABLES PROPERLY.



Connect the cable and secure it using 1 M3*8 screw



The left front leg assembly is completed !

2.2 Assemble the Left Rear Leg

You will need:



1× Left Leg



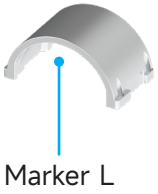
Knee Joint
(KNEE)



Hip Joint
(HIPYLRXX)



Side-swing Joint
(HIPXLRRF)



Leg Hoop（L）



Knee Hoop

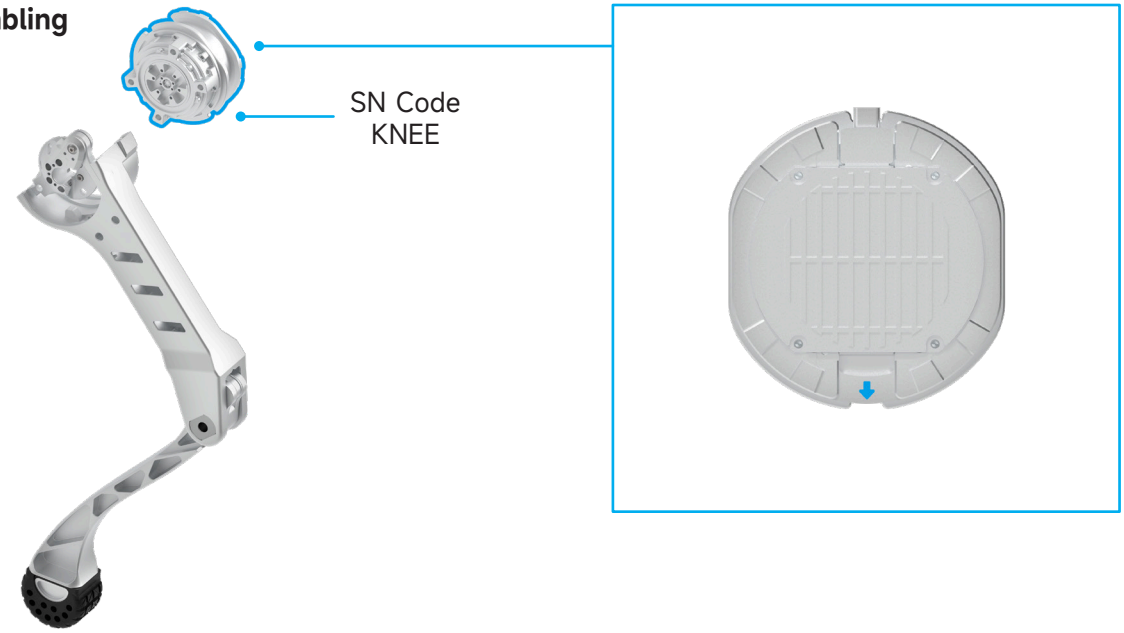
13× M3*8 Screws



2× M6*16 Screws

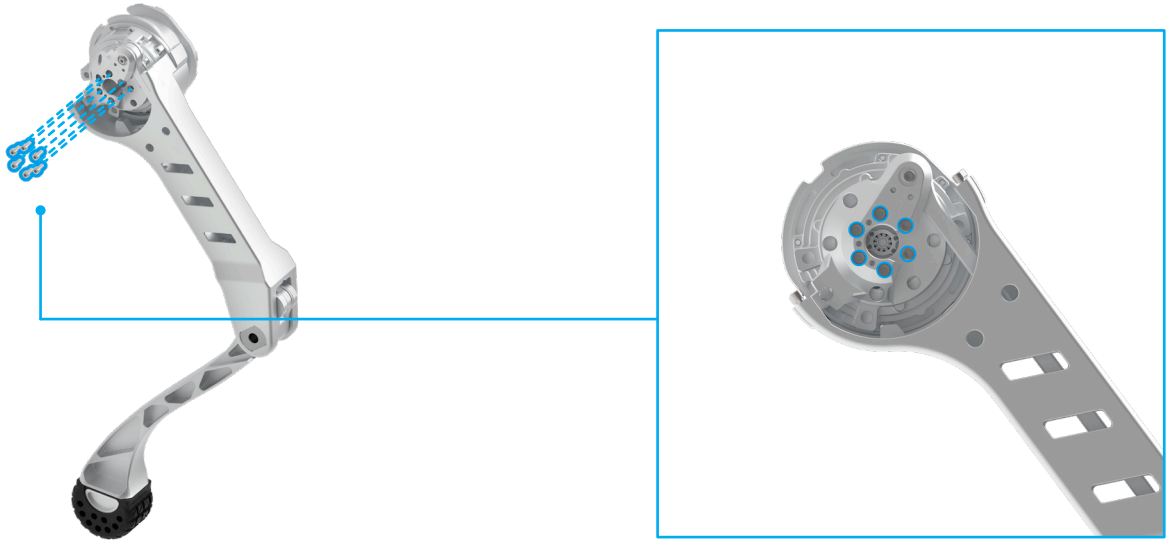
Start Assembling

①



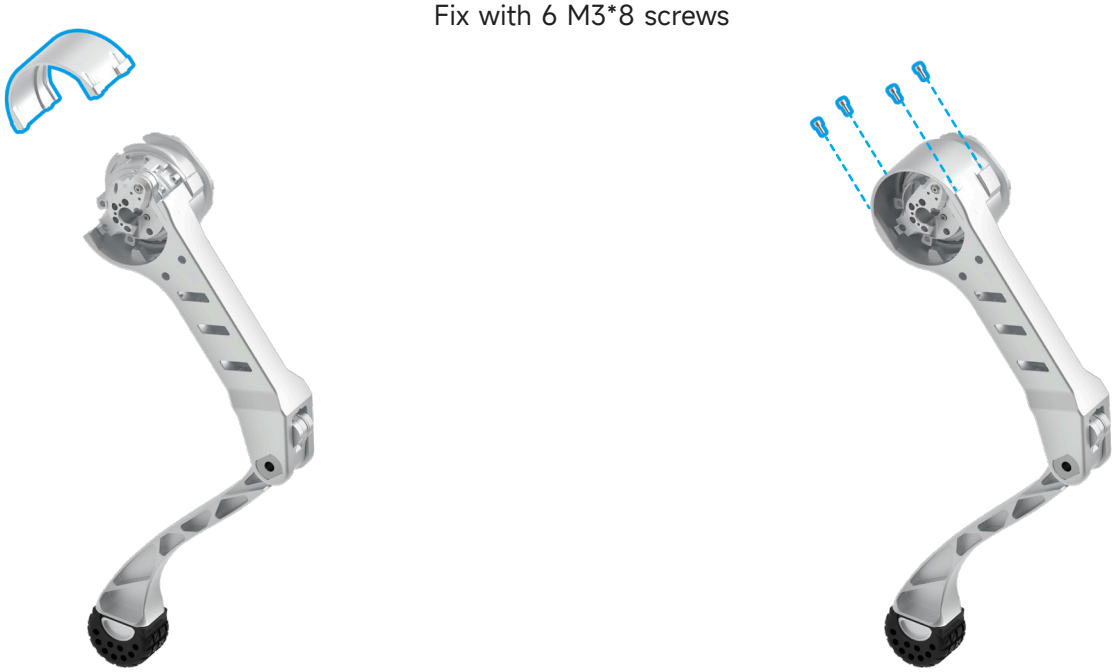
Insert the knee joint into the mounting slot

②



Fix with 6 M3*8 screws

③



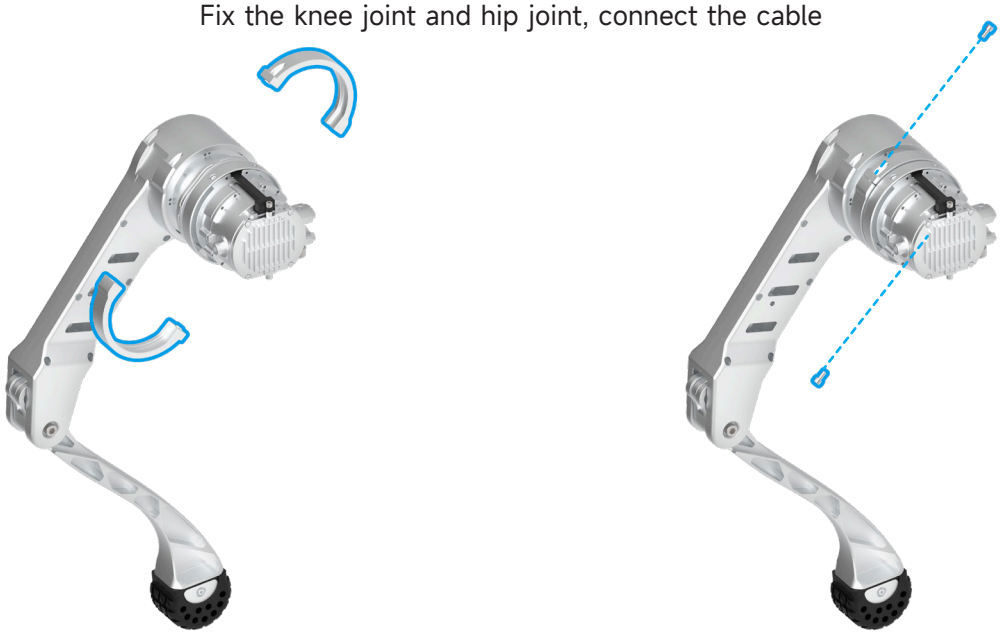
Place the leg hoop over the knee joint and secure it with 4 M3*8 screws

④



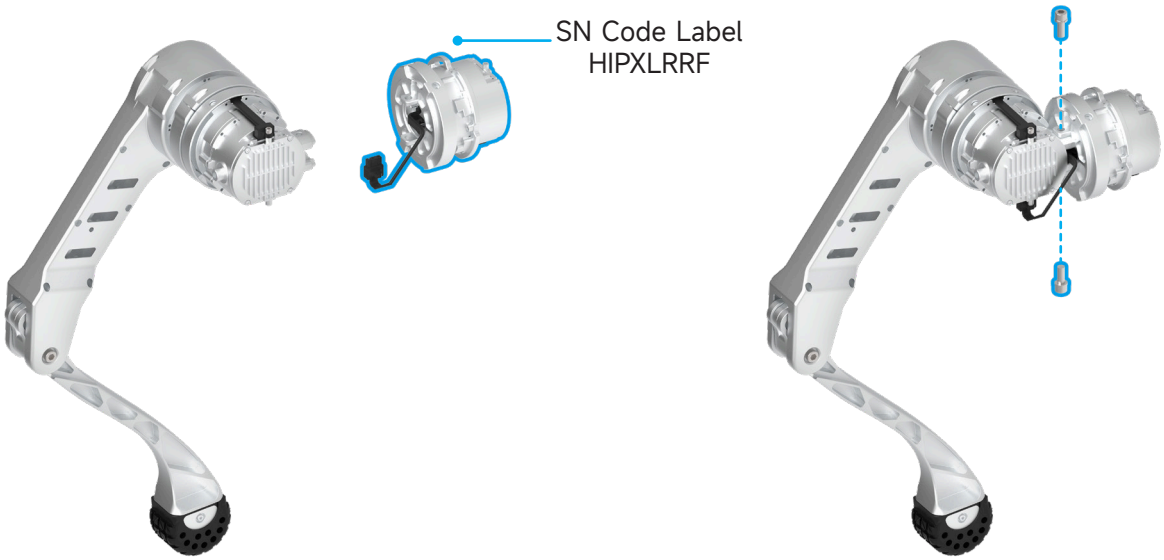
Fix the knee joint and hip joint, connect the cable

⑤



Fix the knee joint and hip joint with 2 knee hoops, and secure them with 2 M3*8 screws.

⑥



Align the side-swing joint mounting hole with the hip joint mounting hole and secure with 2 M6*16 screws
* ROUTE THE MOTOR CABLES PROPERLY.



Connect the cable and secure it using 1 M3*8 screw



The left rear leg assembly is completed !

2.3 Assemble the Right Front Leg

You will need:



1× Right Leg



Knee Joint
(KNEE)



Hip Joint
(HIPYRFXX)



Side-swing Joint
(HIPXLRRF)



Leg Hoop（R）



Knee Hoop

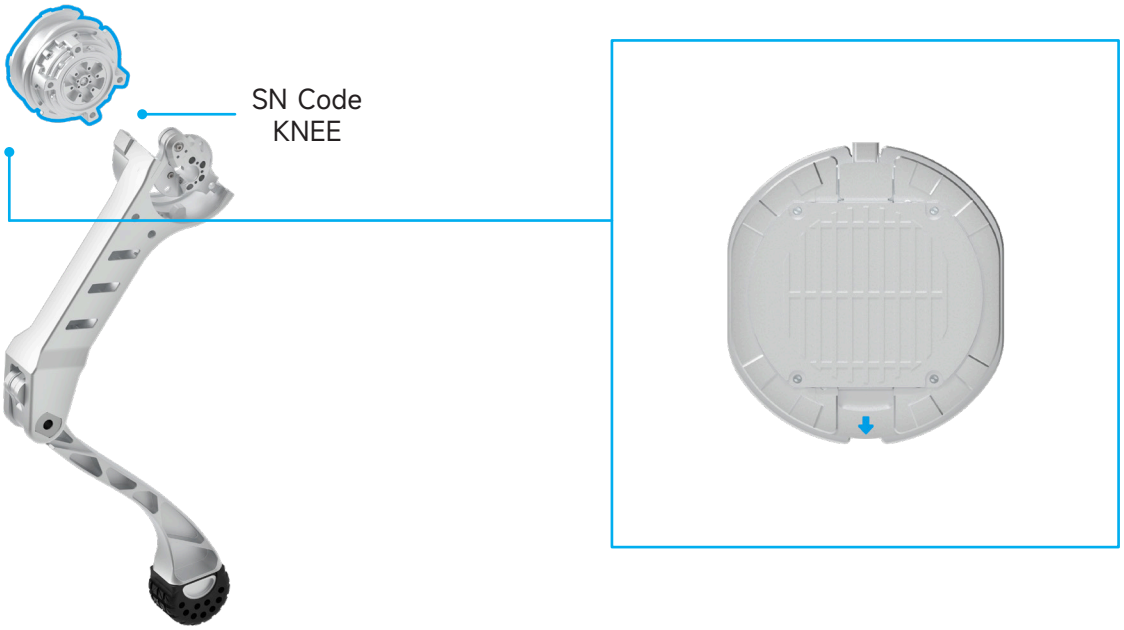
13× M3*8 Screws



2× M6*16 Screws

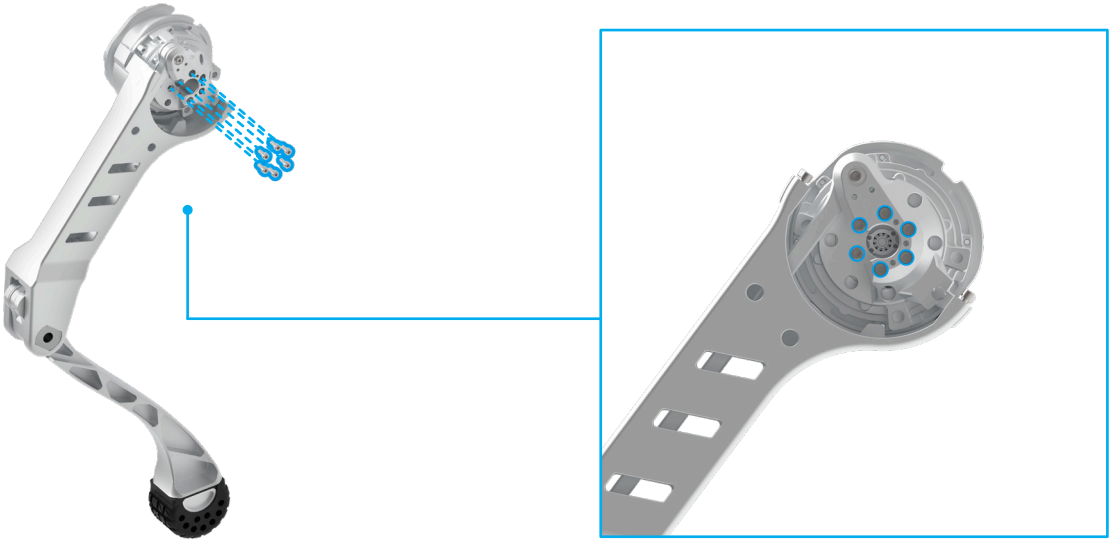
Start Assembling

①



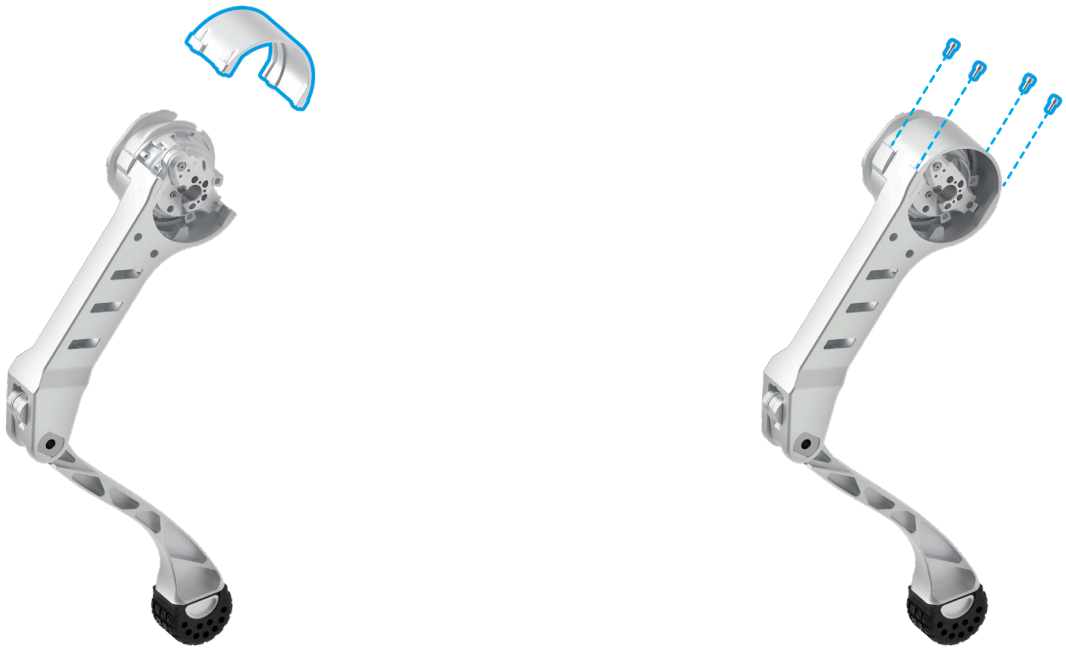
Insert the knee joint into the mounting slot

②



Fix with 6 M3*8 screws

③



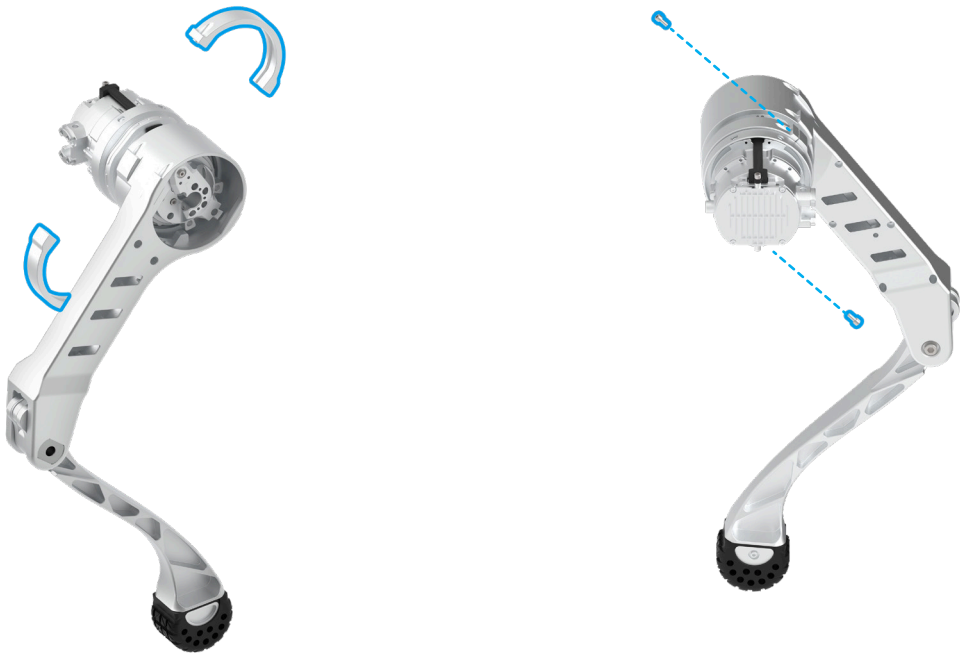
Place the leg hoop over the knee joint and secure it with 4 M3*8 screws.

④



Fix the knee joint and hip joint, connect the cable

⑤

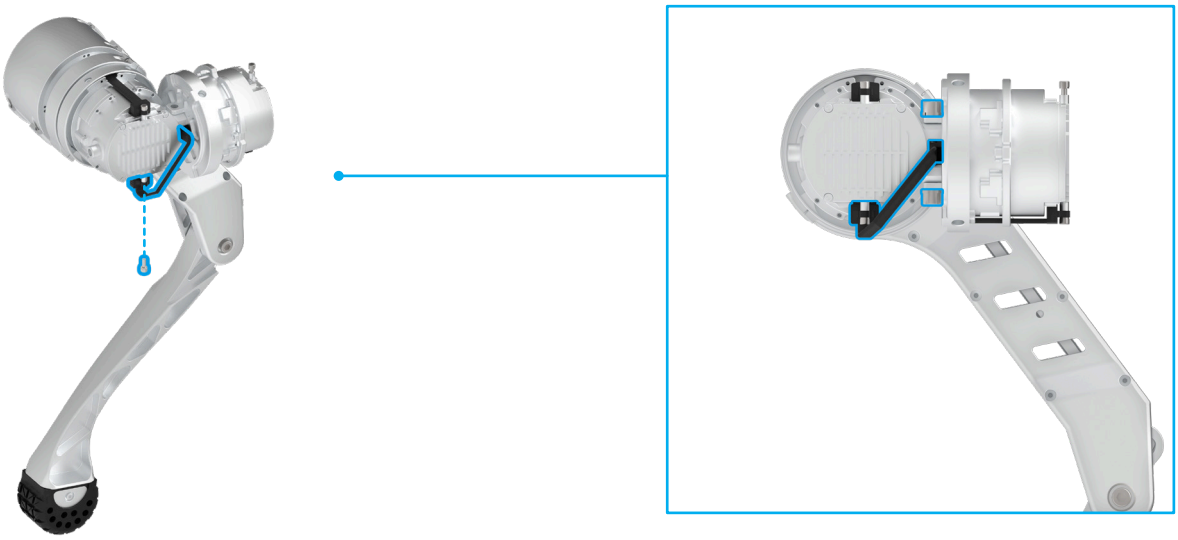


Fix the knee joint and hip joint with 2 knee hoops, and secure them with 2 M3*8 screws.

⑥ SN Code Label
HIPXLRRF



Align the side-swing joint mounting hole with the hip joint mounting hole and secure with 2 M6*16 screws
* ROUTE THE MOTOR CABLES PROPERLY.



Connect the cable and secure it using 1 M3*8 screw



The right front leg assembly is completed !

2.4 Assemble the Right Rear Leg

You will need:



1× Right Leg



Knee Joint
(KNEE)



Hip Joint
(HIPYRRXX)



Side-swing Joint
(HIPXLFRR)



Leg Hoop（R）



Knee Hoop

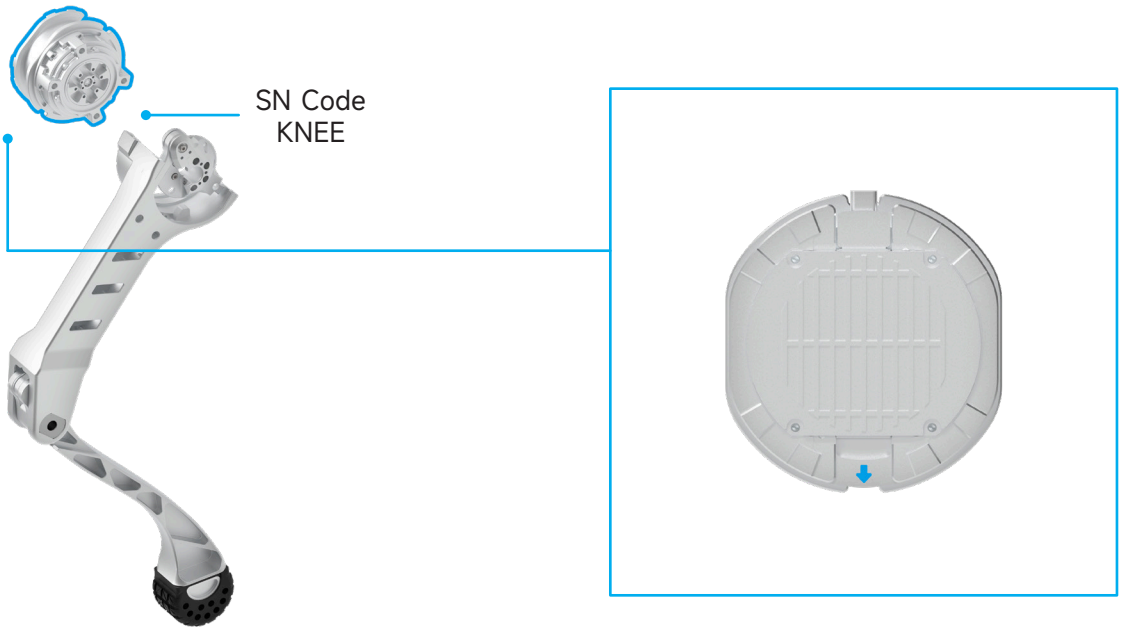
13× M3*8 Screws



2× M6*16 Screws

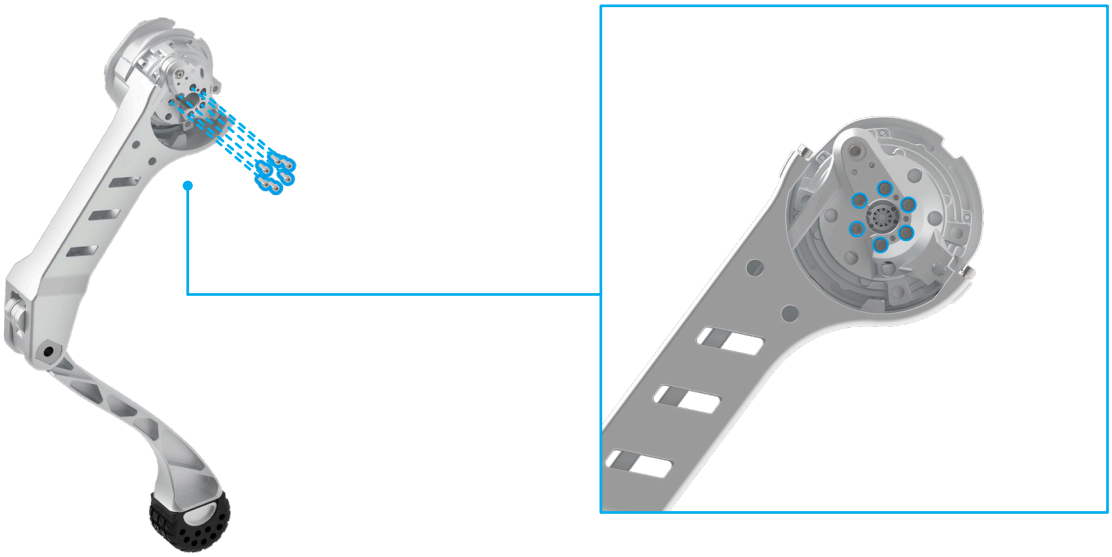
Start Assembling

①



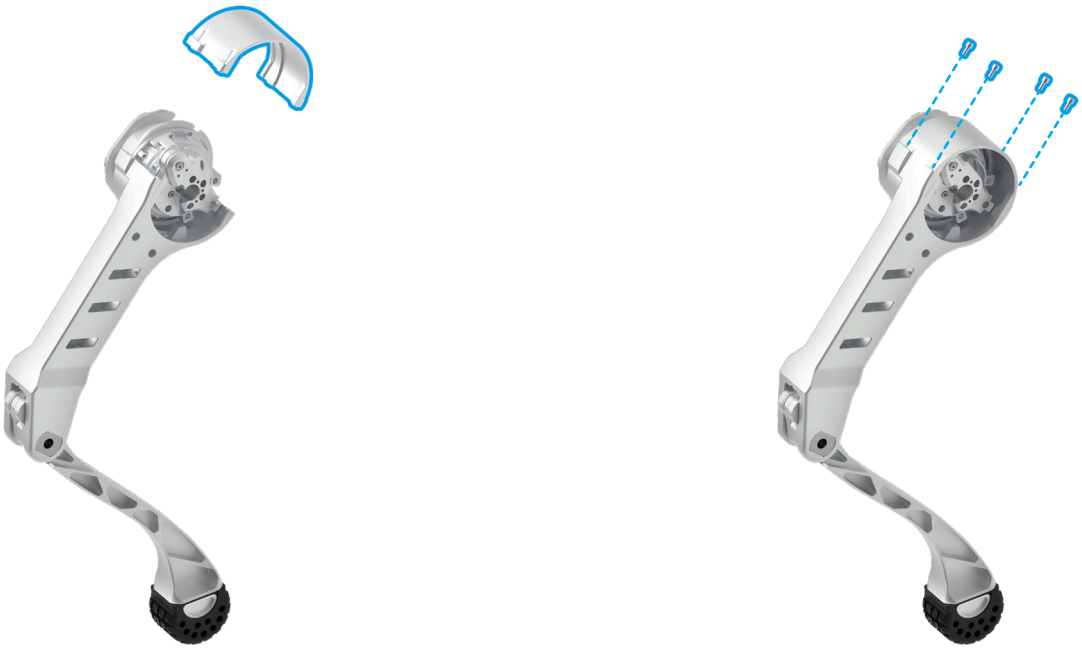
Insert the knee joint into the mounting slot

②



Fix with 6 M3*8 screws

③



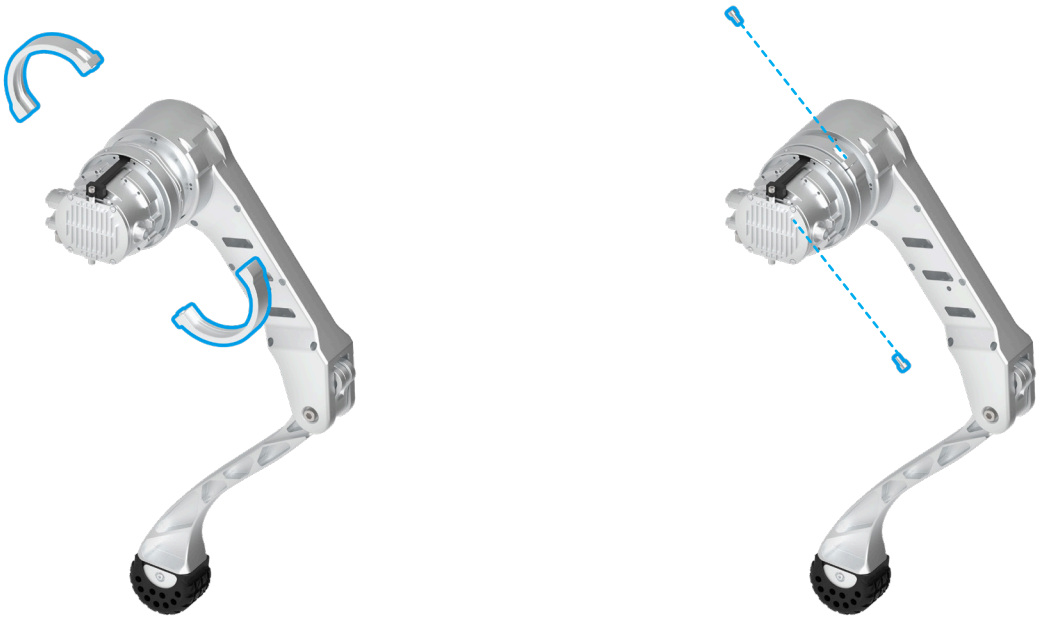
Place the leg hoop over the knee joint and secure it with 4 M3*8 screws

④



Fix the knee joint and hip joint, connect the cable

⑤

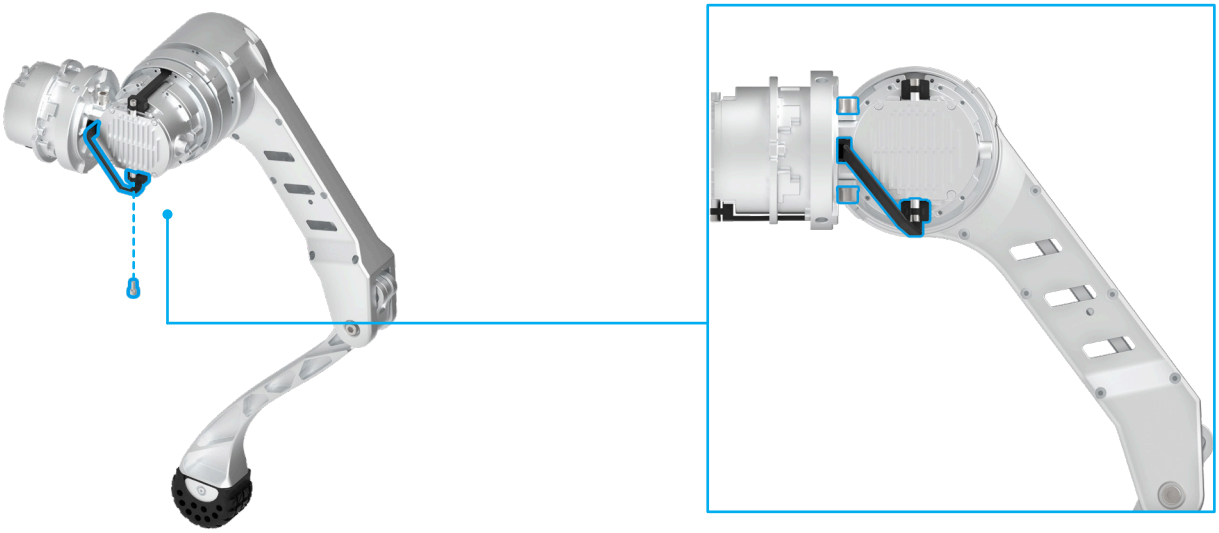


Fix the knee joint and hip joint with 2 knee hoops, and secure them with 2 M3*8 screws.

⑥



Align the side-swing joint mounting hole with the hip joint mounting hole and secure with 2 M6*16 screws
* ROUTE THE MOTOR CABLES PROPERLY.



Connect the cable and secure it using 1 M3*8 screw



The right rear leg assembly is completed !

3. Attach the Four Legs to the Torso

You will need:



Left-front Leg



Left-rear Leg



Right-front Leg



Right-rear Leg



Torso



2× Handles

4× Bushings



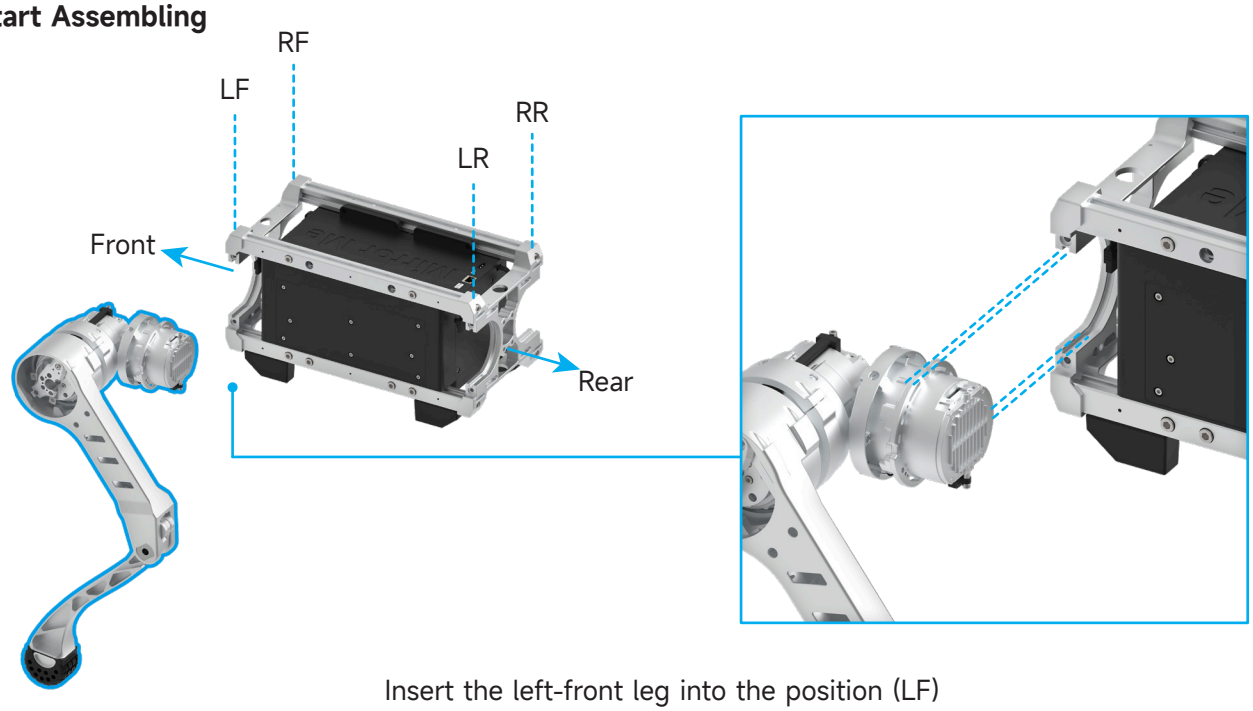
4× Torso Hoops

4× M3*8 Screws

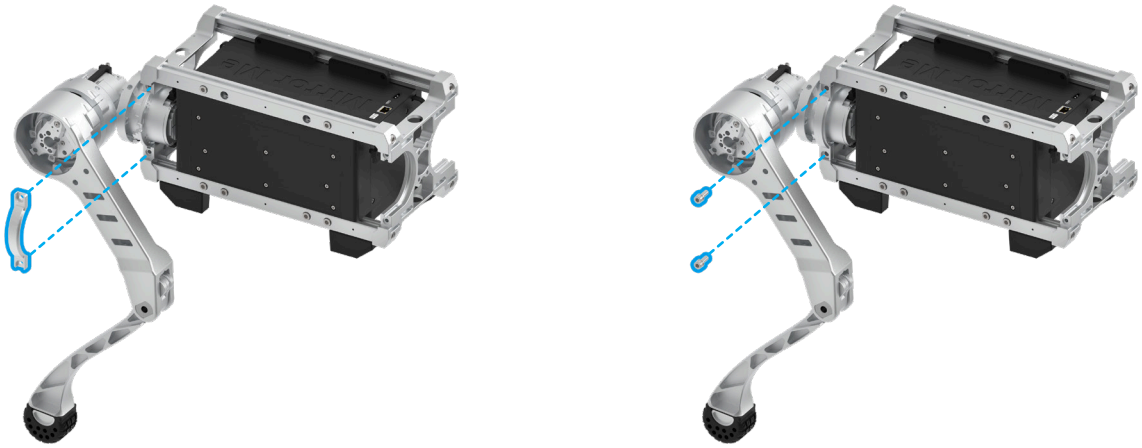


16× M6*16 Screws

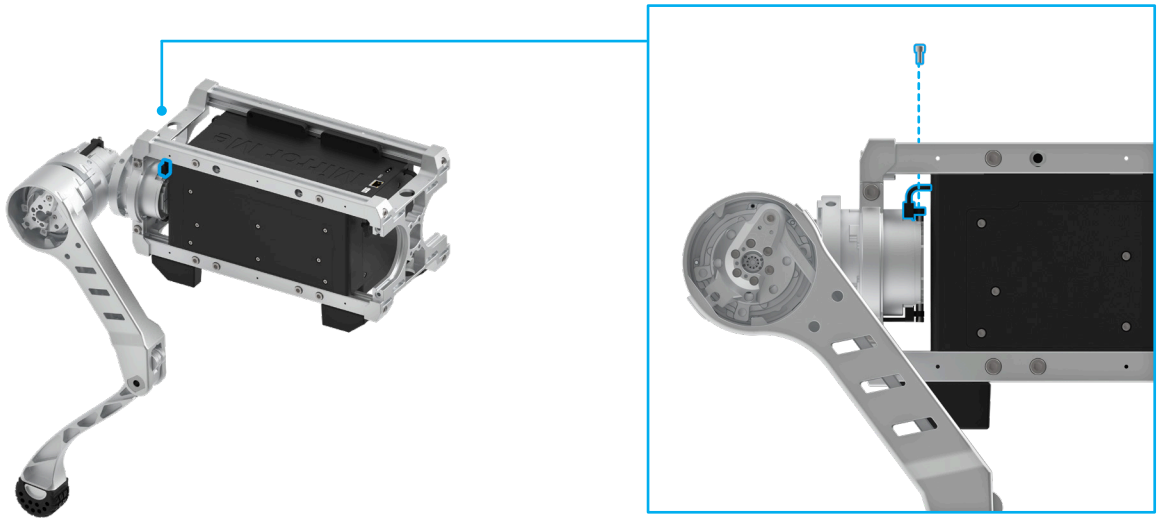
Start Assembling



Insert the left-front leg into the position (LF)



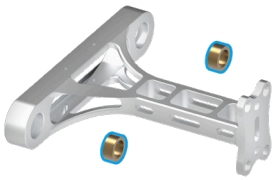
Secure the torso hoop with 2 M6*16 screws



Connect the cable and secure it using 1 M3*8 screw



Install the other three legs in the same way



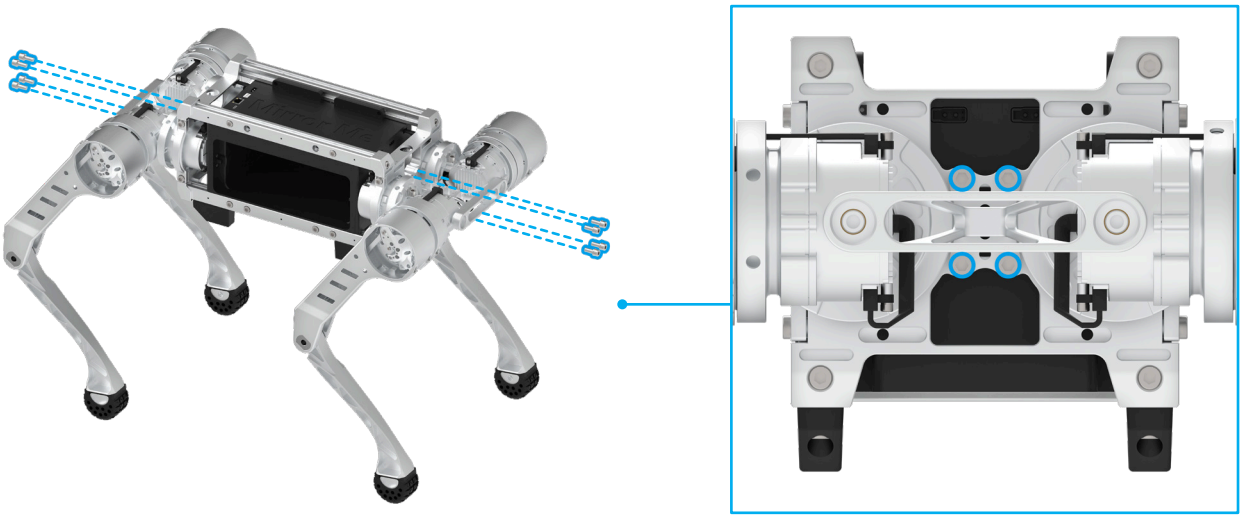


Insert the bushing into the two circular holes of the handle





Locate the screw holes on the fixing base that are used to secure the handles



Secure with 8 M6×16 screws



Accomplished!

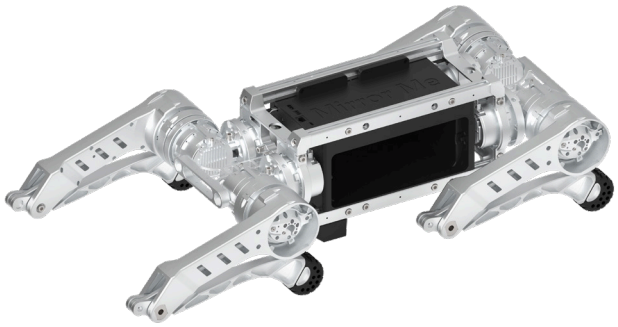
4. Install The Battery

You will need:



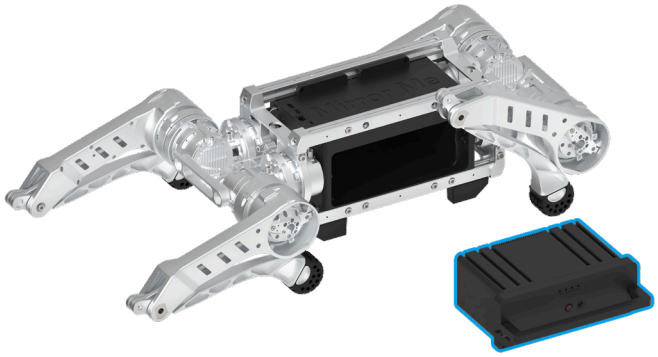
Lithium-ion battery

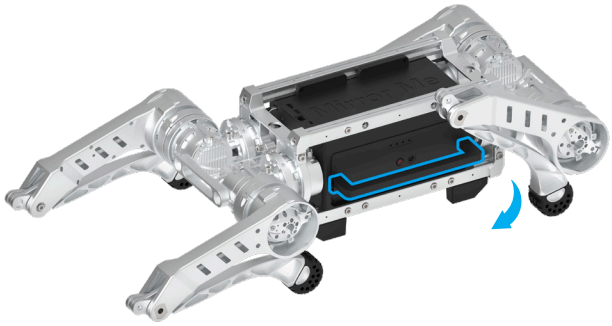
Start Assembling





Position the robotic dog in the "Lie down" posture





Raise the left-front leg, insert the lithium battery, push the battery all the way to the bottom, and return the battery handle to its downward position.



Accomplished!